Calcurion Maths Worksheets

A-Level Mathematics

Topic: Statistics – Binomial Distribution

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. State the conditions for a binomial distribution $X \sim B(n, p)$. **[3]**

2. $X \sim B(5, 0.4)$. Find $P(X = 2)$. **[3]**

3. $X \sim B(6, 0.3)$. Find $P(X = 0)$. **[2]**

4. $X \sim B(8, 0.25)$. Find $P(X = 3)$. **[3]**

5. $X \sim B(10, 0.5)$. Find $P(X \leq 2)$. **[4]**

6. $X \sim B(4, 0.6)$. Find $E(X)$ and $\text{Var}(X)$. **[3]**

7. $X \sim B(12, 0.4)$. Find the mean and standard deviation. **[3]**

8. A fair coin is tossed 8 times. Find the probability of exactly 5 heads. **[3]**

9. A die is rolled 5 times. Find $P(\text{exactly 2 sixes})$. **[3]**

10. $X \sim B(7, 0.2)$. Find $P(X \geq 1)$. **[3]**

11. $X \sim B(9, 0.35)$. Find $P(X = 4)$. **[3]**

12. The probability a component is defective is 0.1. In a batch of 15, find $P(\text{exactly 2 defective})$. **[3]**

13. $X \sim B(20, 0.3)$. Write an expression for $P(X = 5)$ (do not evaluate). **[2]**

14. If $X \sim B(n, p)$ and $E(X) = 12$, $\text{Var}(X) = 3$, find n and p. **[4]**

15. $X \sim B(5, 0.5)$. Find the mode. **[2]**

Section B – Hard (Q16–25)

16. $X \sim B(10, 0.4)$. Find $P(3 \leq X \leq 6)$. **[5]**

17. A shooter hits the target with probability 0.7 independently each shot. In 12 shots, find $P(\text{at least 10 hits})$. **[5]**

18. $X \sim B(15, 0.2)$. Find $P(X \leq 2)$ using cumulative tables or calculation. **[4]**

19. The probability a seed germinates is 0.85. In a tray of 20 seeds, find the expected number that germinate and $P(\text{all germinate})$. **[4]**

20. $X \sim B(n, 0.25)$ and $E(X) = 5$. Find n and $P(X = 2)$. **[4]**

21. A multiple-choice test has 10 questions each with 4 options. A student guesses all. Find $P(\text{exactly 3 correct})$. **[3]**

22. $X \sim B(8, p)$ and $P(X = 0) = 0.01$. Find p and then $E(X)$. **[5]**

23. Explain why a binomial model may not be appropriate for the number of accidents at a junction in a week. **[3]**

24. $X \sim B(12, 0.3)$. Find the values of a and b such that $P(X \leq a) \leq 0.1$ and $P(X \geq b) \leq 0.1$ (critical regions style). **[5]**

25. Two independent binomial random variables $X \sim B(5, 0.4)$, $Y \sim B(5, 0.4)$. Find $P(X + Y = 3)$. **[5]**

End of Worksheet • *Show all working clearly.*